Case Study 2: Global Conflict Analysis

Project Title
Global Conflict & Terrorism – EDA, SQL Analysis & Dashboard

Project Objective
To analyse global conflict data from 1990–2023, identify regional hotspots, understand
attack patterns, and track how conflict has evolved over decades using Python, SQL and
Power BI.

Tools & Technologies Used
1. **Python (Pandas, NumPy)** – Data cleaning, feature engineering, EDA
2. **Seaborn & Matplotlib** – Distribution and trend visualizations
3. **Folium** – Interactive geographic heatmap of conflict locations
4. **SQL** – Querying conflict patterns, casualty rankings, success rates
5. **Power BI** – Regional and yearly conflict dashboard

Data Overview
- 1,390+ conflict incidents across 1990–2023
- Key attributes: Year, Region, Country, Attack Type, Group Name, Killed, Wounded,
  Total Casualties, Target Type, Weapon Type, Success, Latitude, Longitude

Analysis Performed

Univariate Analysis
- Middle East and Africa have the highest incident counts
- Bombing/Explosion is the most frequent attack type

Bivariate Analysis
- Incidents and casualties peaked between 2010–2017
- Bombing causes the highest average deaths per incident
- Hijacking and Assassination have the lowest success rates

Multivariate Analysis

- Region × Decade heatmap shows Middle East spiking in the 2010s
- Africa shows consistently high activity across all decades

SQL Analysis
- Top 10 countries by total casualties
- Success rate by attack type and weapon type
- Deadliest years ranked by total killed
- Decade-wise casualty trends

Geographic Map
- Folium heatmap plotted using latitude/longitude — visual hotspot clusters clearly visible in Middle East, South Asia and parts of Africa

Key Insights
- Middle East and Africa account for the majority of global conflict incidents
- Conflict peaked globally between 2010–2017 and declined post-2018
- Bombing is the deadliest and most common attack method
- Overall attack success rate is ~73% — most executed attacks succeed
- Civilians are the most frequent target type across all regions

Business Outcomes
- Policy makers can identify highest-risk regions and decades
- Security analysts can prioritise attack types for prevention strategies
- Dashboard enables year-by-year and region-by-region drill-down

What I Learned
- Writing analytical SQL queries on real-world event data
- Geographic visualisation using Folium heatmaps
- Decade-level trend analysis using pivot tables
- Combining Python + SQL + Power BI in one end-to-end project

POWER BI:

MY SQL

use global_conflict
select * from global_conflict_analysis

-- 1. Which region has the most conflict incidents?
SELECT region, COUNT(*) AS total_incidents
FROM global_conflict_analysis
GROUP BY region
ORDER BY total_incidents DESC;

-- 2. Which country has the highest total casualties?
SELECT country, SUM(total_casualties) AS total_casualties
FROM global_conflict_analysis
GROUP BY country
ORDER BY total_casualties DESC
LIMIT 10;

```
-- 3. Which attack type is most common?
SELECT attack_type, COUNT(*) AS count
FROM global_conflict_analysis
GROUP BY attack_type
ORDER BY count DESC;

-- 4. Which attack type kills the most people on average?
SELECT attack_type, ROUND(AVG(killed), 2) AS avg_killed
FROM global_conflict_analysis
GROUP BY attack_type
ORDER BY avg_killed DESC;

-- 5. How has number of incidents changed per year?
SELECT year, COUNT(*) AS incidents
FROM global_conflict_analysis
GROUP BY year
ORDER BY year;

-- 6. Which decade had the most casualties?
SELECT (year/10)*10 AS decade, SUM(total_casualties) AS total_casualties
FROM global_conflict_analysis
GROUP BY decade
ORDER BY decade;

-- 7. What is the overall attack success rate?
SELECT ROUND(AVG(success) * 100, 2) AS success_rate_percent
FROM global_conflict_analysis;

-- 8. Which attack type has the highest success rate?
SELECT attack_type, ROUND(AVG(success) * 100, 2) AS success_rate
FROM global_conflict_analysis
GROUP BY attack_type
ORDER BY success_rate DESC;

-- 9. Which weapon type causes the most casualties?
SELECT weapon_type, ROUND(AVG(total_casualties), 2) AS avg_casualties
```

```sql
FROM global_conflict_analysis
GROUP BY weapon_type
ORDER BY avg_casualties DESC;

-- 10. Which group is most active? (most incidents)
SELECT group_name, COUNT(*) AS incidents
FROM global_conflict_analysis
GROUP BY group_name
ORDER BY incidents DESC
LIMIT 5;

-- 11. Which target type suffers the most casualties?
SELECT target_type, SUM(killed) AS total_killed
FROM global_conflict_analysis
GROUP BY target_type
ORDER BY total_killed DESC;

-- 12. Most dangerous year overall?
SELECT year, SUM(killed) AS total_killed, COUNT(*) AS incidents
FROM global_conflict_analysis
GROUP BY year
ORDER BY total_killed DESC
LIMIT 5;
```